

The Chain Rule

ANSWER KEY



Basic chain rule applications

1 Differentiate each function using the chain rule:

a $f(x) = (4x - 1)^5$
 $= 5(4x - 1)^4(4) = 20(4x - 1)^4$

c $f(x) = e^{5x} = 5e^{5x}$

b $f(x) = \sin(3x^2) = 6x\cos(3x^2)$

d $f(x) = \ln(2x + 7) = \frac{2}{2x + 7}$

2 Differentiate $f(x) = \sqrt{x^2 + 9}$.

$$f'(x) = \frac{2x}{2\sqrt{x^2 + 9}} = \frac{x}{\sqrt{x^2 + 9}}.$$

The chain rule with trig and exponential functions

3 Differentiate $f(x) = \cos^3(x)$ (i.e. $[\cos x]^3$).

$$f'(x) = 3\cos^2 x \cdot (-\sin x) = -3\cos^2 x \sin x.$$

4 Differentiate $f(x) = e^{\sin x}$.

$$f'(x) = \cos x \cdot e^{\sin x}.$$

5 Differentiate $f(x) = \tan(e^x)$.

$$f'(x) = \sec^2(e^x) \cdot e^x.$$

Nested (multiple) chain rule

6 Differentiate $f(x) = \sin(\sqrt{x^2 + 1})$, applying the chain rule twice.

$$f'(x) = \cos(\sqrt{x^2 + 1}) \cdot \frac{x}{\sqrt{x^2 + 1}}.$$

7 Differentiate $f(x) = (\ln(3x))^4$.

$$f'(x) = 4(\ln(3x))^3 \cdot \frac{3}{3x} = \frac{4(\ln(3x))^3}{x}.$$

Chain rule combined with product/quotient rules

8 Differentiate $f(x) = x^2 e^{3x}$.

$$f'(x) = 2xe^{3x} + x^2(3e^{3x}) = e^{3x}(2x + 3x^2).$$

- 9 Differentiate $f(x) = \frac{\sin(2x)}{x}$.

$$f'(x) = \frac{2\cos(2x) \cdot x - \sin(2x)}{x^2}.$$

Chain rule with tables of values

- 10 Given $f(2) = 5$, $f'(2) = 3$, $g(2) = 7$, $g'(2) = -1$, find $\frac{d}{dx}[f(g(x))]$ at $x = 2$ if $g(2) = 7$ and $f'(7) = 4$ (note the composition uses f' evaluated at $g(2) = 7$, not at 2).

$$\frac{d}{dx}f(g(x)) = f'(g(x)) \cdot g'(x); \text{ at } x = 2: f'(7) \cdot g'(2) = 4 \cdot (-1) = -4.$$

Chain rule with inverse trig and logarithms

- 11 Differentiate $f(x) = \sin^{-1}(2x)$.

$$f'(x) = \frac{2}{\sqrt{1-(2x)^2}} = \frac{2}{\sqrt{1-4x^2}}.$$

- 12 Differentiate $f(x) = \ln(\sin x)$.

$$f'(x) = \frac{\cos x}{\sin x} = \cot x.$$

- 13 Differentiate $f(x) = \tan^{-1}(x^2)$.

$$f'(x) = \frac{2x}{1+(x^2)^2} = \frac{2x}{1+x^4}.$$

Finding tangent lines using the chain rule

- 14 Find the equation of the tangent line to $f(x) = (2x - 1)^3$ at $x = 1$.

$$f(1) = 1^3 = 1. \quad f'(x) = 3(2x - 1)^2(2) = 6(2x - 1)^2; \quad f'(1) = 6(1)^2 = 6. \quad \text{Tangent: } y - 1 = 6(x - 1), \text{ i.e. } y = 6x - 5.$$

Visualizing a composite function and its derivative

- 15 The graph shows $f(x) = (x - 1)^2$ (an inner-outer composite of $u = x - 1$ and u^2), with its tangent line at $x = 2$.

a Use the chain rule to find $f'(x)$. $f'(x) = 2(x - 1)(1) = 2(x - 1)$.

b Confirm $f'(2)$ matches the slope of the tangent line shown. $f'(2) = 2(1) = 2$, matching the tangent line $y = 2x - 2$, which has slope 2.

The chain rule in applied rate problems

- 16** The volume of a sphere is $V = \frac{4}{3}\pi r^3$, and the radius is growing according to $r(t) = 2t + 1$ (cm, seconds). Use the chain rule to find $\frac{dV}{dt}$ at $t = 2$.

$$\frac{dV}{dt} = \frac{dV}{dr} \cdot \frac{dr}{dt} = 4\pi r^2 \cdot 2. \text{ At } t = 2: r = 5, \text{ so } \frac{dV}{dt} = 4\pi(25)(2) = 200\pi \approx 628.3 \text{ cm}^3/\text{s}.$$

Chain rule with a table of composed values

- 17** Given $h(x) = f(g(x))$ where $g(1) = 3$, $g'(1) = 2$, $f(3) = 7$, $f'(3) = -4$, find $h'(1)$.

$$h'(x) = f'(g(x)) \cdot g'(x): h'(1) = f'(g(1)) \cdot g'(1) = f'(3) \cdot 2 = (-4)(2) = -8.$$

- 18** Differentiate $f(x) = \sqrt[3]{x^2 + 1}$ using the chain rule.

$$f(x) = (x^2 + 1)^{1/3}: f'(x) = \frac{1}{3}(x^2 + 1)^{-2/3}(2x) = \frac{2x}{3(x^2 + 1)^{2/3}}.$$

Multiple choice — easy 15

- 19** Use the chain rule to differentiate $f(x) = (x - 1)^2$, then find $f'(1)$.

a. 0 ✓

b. -2

c. 3

d. 5

$$f'(x) = 2(x - 1)^1(1), \text{ so } f'(1) = 2(0)^1(1) = 0.$$

- 20** Use the chain rule to differentiate $f(x) = (x - 2)^3$, then find $f'(0)$.

a. 12 ✓

b. 4

c. 10

d. -24

$$f'(x) = 3(x - 2)^2(1), \text{ so } f'(0) = 3(-2)^2(1) = 12.$$

21 Use the chain rule to differentiate $f(x) = (4x - 3)^2$, then find $f'(1)$.

a. 4

b. 6

c. 2

d. 8 ✓

$$f'(x) = 2(4x - 3)^1(4), \text{ so } f'(1) = 2(1)^1(4) = 8.$$

22 Use the chain rule to differentiate $f(x) = (2x - 2)^4$, then find $f'(1)$.

a. -2

b. 3

c. 0 ✓

d. 2

$$f'(x) = 4(2x - 2)^3(2), \text{ so } f'(1) = 4(0)^3(2) = 0.$$

23 Use the chain rule to differentiate $f(x) = (4x - 1)^2$, then find $f'(1)$.

a. 24 ✓

b. 72

c. 12

d. 6

$$f'(x) = 2(4x - 1)^1(4), \text{ so } f'(1) = 2(3)^1(4) = 24.$$

24 Use the chain rule to differentiate $f(x) = (3x + 1)^3$, then find $f'(0)$.

a. 14

b. 12

c. 3

d. 9 ✓

$$f'(x) = 3(3x + 1)^2(3), \text{ so } f'(0) = 3(1)^2(3) = 9.$$

25 Use the chain rule to differentiate $f(x) = (3x)^4$, then find $f'(1)$.

a. 81

b. 324 ✓

c. 108

d. 972

$f'(x) = 4(3x)^3(3)$, so $f'(1) = 4(3)^3(3) = 324$.

26 Use the chain rule to differentiate $f(x) = (2x + 4)^4$, then find $f'(-2)$.

a. 5

b. 3

c. -2

d. 0 ✓

$f'(x) = 4(2x + 4)^3(2)$, so $f'(-2) = 4(0)^3(2) = 0$.

27 Use the chain rule to differentiate $f(x) = (2x + 3)^4$, then find $f'(0)$.

a. 108

b. 648

c. 216 ✓

d. 54

$f'(x) = 4(2x + 3)^3(2)$, so $f'(0) = 4(3)^3(2) = 216$.

28 Use the chain rule to differentiate $f(x) = (x + 3)^4$, then find $f'(0)$.

- a. 27
- b. 324
- c. 110
- d. 108 ✓

$$f'(x) = 4(x + 3)^3(1), \text{ so } f'(0) = 4(3)^3(1) = 108.$$

29 Use the chain rule to differentiate $f(x) = (x)^3$, then find $f'(1)$.

- a. 5
- b. 1
- c. 0
- d. 3 ✓

$$f'(x) = 3(x)^2(1), \text{ so } f'(1) = 3(1)^2(1) = 3.$$

30 Use the chain rule to differentiate $f(x) = (x - 1)^3$, then find $f'(0)$.

- a. 3 ✓
- b. -3
- c. 0
- d. 1

$$f'(x) = 3(x - 1)^2(1), \text{ so } f'(0) = 3(-1)^2(1) = 3.$$

31 Use the chain rule to differentiate $f(x) = (4x + 6)^2$, then find $f'(-2)$.

- a. -16 ✓
- b. -8
- c. -4
- d. 32

$$f'(x) = 2(4x + 6)^1(4), \text{ so } f'(-2) = 2(-2)^1(4) = -16.$$

32 Use the chain rule to differentiate $f(x) = (2x + 4)^2$, then find $f'(-2)$.

a. -2

b. 0 ✓

c. 2

d. 5

$$f'(x) = 2(2x + 4)^1(2), \text{ so } f'(-2) = 2(0)^1(2) = 0.$$

33 Use the chain rule to differentiate $f(x) = (x + 1)^4$, then find $f'(1)$.

a. 64

b. 8

c. 32 ✓

d. 35

$$f'(x) = 4(x + 1)^3(1), \text{ so } f'(1) = 4(2)^3(1) = 32.$$

Multiple choice — medium 15

34 Given $g(5) = 4$, $g'(5) = -2$, and $f'(4) = 5$, find $h'(5)$ if $h(x) = f(g(x))$.

a. 5

b. -2

c. 3

d. -10 ✓

$$h'(x) = f'(g(x)) \cdot g'(x), \text{ so } h'(5) = f'(g(5)) \cdot g'(5) = f'(4) \cdot (-2) = (5)(-2) = -10.$$

35 Given $g(7) = -5$, $g'(7) = -5$, and $f'(-5) = 2$, find $h'(7)$ if $h(x) = f(g(x))$.

- a. -5
- b. -3
- c. -10 ✓
- d. 2

$$h'(x) = f'(g(x)) \cdot g'(x), \text{ so } h'(7) = f'(g(7)) \cdot g'(7) = f'(-5) \cdot (-5) = (2)(-5) = -10.$$

36 Given $g(6) = -5$, $g'(6) = -6$, and $f'(-5) = -4$, find $h'(6)$ if $h(x) = f(g(x))$.

- a. -6
- b. 24 ✓
- c. -4
- d. -10

$$h'(x) = f'(g(x)) \cdot g'(x), \text{ so } h'(6) = f'(g(6)) \cdot g'(6) = f'(-5) \cdot (-6) = (-4)(-6) = 24.$$

37 Given $g(5) = 0$, $g'(5) = -2$, and $f'(0) = -6$, find $h'(5)$ if $h(x) = f(g(x))$.

- a. 12 ✓
- b. -8
- c. -6
- d. -2

$$h'(x) = f'(g(x)) \cdot g'(x), \text{ so } h'(5) = f'(g(5)) \cdot g'(5) = f'(0) \cdot (-2) = (-6)(-2) = 12.$$

38 Given $g(4) = -3$, $g'(4) = 6$, and $f'(-3) = 3$, find $h'(4)$ if $h(x) = f(g(x))$.

- a. 9
- b. 18 ✓
- c. 6
- d. 3

$$h'(x) = f'(g(x)) \cdot g'(x), \text{ so } h'(4) = f'(g(4)) \cdot g'(4) = f'(-3) \cdot (6) = (3)(6) = 18.$$

39 Given $g(7) = 3$, $g'(7) = 1$, and $f'(3) = 4$, find $h'(7)$ if $h(x) = f(g(x))$.

a. 1

b. 9

c. 4 ✓

d. 5

$$h'(x) = f'(g(x)) \cdot g'(x), \text{ so } h'(7) = f'(g(7)) \cdot g'(7) = f'(3) \cdot (1) = (4)(1) = 4.$$

40 Given $g(3) = -3$, $g'(3) = 6$, and $f'(-3) = -4$, find $h'(3)$ if $h(x) = f(g(x))$.

a. -24 ✓

b. -4

c. 6

d. 2

$$h'(x) = f'(g(x)) \cdot g'(x), \text{ so } h'(3) = f'(g(3)) \cdot g'(3) = f'(-3) \cdot (6) = (-4)(6) = -24.$$

41 Given $g(7) = -4$, $g'(7) = 3$, and $f'(-4) = 5$, find $h'(7)$ if $h(x) = f(g(x))$.

a. 8

b. 5

c. 3

d. 15 ✓

$$h'(x) = f'(g(x)) \cdot g'(x), \text{ so } h'(7) = f'(g(7)) \cdot g'(7) = f'(-4) \cdot (3) = (5)(3) = 15.$$

42 Given $g(8) = 6$, $g'(8) = -3$, and $f'(6) = -3$, find $h'(8)$ if $h(x) = f(g(x))$.

a. -6

b. 14

c. -3

d. 9 ✓

$$h'(x) = f'(g(x)) \cdot g'(x), \text{ so } h'(8) = f'(g(8)) \cdot g'(8) = f'(6) \cdot (-3) = (-3)(-3) = 9.$$

43 Given $g(3) = -5$, $g'(3) = 6$, and $f'(-5) = 2$, find $h'(3)$ if $h(x) = f(g(x))$.

a. 2

b. 6

c. 12 ✓

d. 8

$$h'(x) = f'(g(x)) \cdot g'(x), \text{ so } h'(3) = f'(g(3)) \cdot g'(3) = f'(-5) \cdot (6) = (2)(6) = 12.$$

44 Given $g(6) = -5$, $g'(6) = -3$, and $f'(-5) = 6$, find $h'(6)$ if $h(x) = f(g(x))$.

a. -18 ✓

b. -3

c. 3

d. 6

$$h'(x) = f'(g(x)) \cdot g'(x), \text{ so } h'(6) = f'(g(6)) \cdot g'(6) = f'(-5) \cdot (-3) = (6)(-3) = -18.$$

45 Given $g(2) = -5$, $g'(2) = 1$, and $f'(-5) = -5$, find $h'(2)$ if $h(x) = f(g(x))$.

a. -5 ✓

b. -4

c. -9

d. 1

$$h'(x) = f'(g(x)) \cdot g'(x), \text{ so } h'(2) = f'(g(2)) \cdot g'(2) = f'(-5) \cdot (1) = (-5)(1) = -5.$$

46 Given $g(3) = 1$, $g'(3) = -1$, and $f'(1) = -4$, find $h'(3)$ if $h(x) = f(g(x))$.

a. -5

b. 4 ✓

c. -1

d. -4

$$h'(x) = f'(g(x)) \cdot g'(x), \text{ so } h'(3) = f'(g(3)) \cdot g'(3) = f'(1) \cdot (-1) = (-4)(-1) = 4.$$

47 Given $g(3) = -6$, $g'(3) = -2$, and $f'(-6) = -3$, find $h'(3)$ if $h(x) = f(g(x))$.

a. -5

b. -3

c. 6 ✓

d. -2

$$h'(x) = f'(g(x)) \cdot g'(x), \text{ so } h'(3) = f'(g(3)) \cdot g'(3) = f'(-6) \cdot (-2) = (-3)(-2) = 6.$$

48 Given $g(6) = 6$, $g'(6) = 3$, and $f'(6) = -6$, find $h'(6)$ if $h(x) = f(g(x))$.

a. 3

b. -3

c. -6

d. -18 ✓

$$h'(x) = f'(g(x)) \cdot g'(x), \text{ so } h'(6) = f'(g(6)) \cdot g'(6) = f'(6) \cdot (3) = (-6)(3) = -18.$$

Multiple choice — hard 15

49 Use the chain rule to differentiate $f(x) = \sqrt{x^2 + 16}$, then find $f'(-3)$.

- a. $-\frac{5}{3}$
- b. $-\frac{3}{4}$
- c. $-\frac{3}{5}$ ✓
- d. $\frac{3}{5}$

$$f'(x) = \frac{x}{\sqrt{x^2 + 16}}, \text{ so } f'(-3) = \frac{-3}{\sqrt{(-3)^2 + 16}} = \frac{-3}{5} = -\frac{3}{5}.$$

50 Use the chain rule to differentiate $f(x) = \sqrt{x^2 + 144}$, then find $f'(5)$.

- a. $\frac{13}{5}$
- b. $-\frac{5}{13}$
- c. $\frac{5}{13}$ ✓
- d. $\frac{5}{12}$

$$f'(x) = \frac{x}{\sqrt{x^2 + 144}}, \text{ so } f'(5) = \frac{5}{\sqrt{(5)^2 + 144}} = \frac{5}{13} = \frac{5}{13}.$$

51 Use the chain rule to differentiate $f(x) = \sqrt{x^2 + 144}$, then find $f'(-5)$.

- a. $-\frac{5}{12}$
- b. $-\frac{13}{5}$
- c. $-\frac{5}{13}$ ✓
- d. $\frac{5}{13}$

$$f'(x) = \frac{x}{\sqrt{x^2 + 144}}, \text{ so } f'(-5) = \frac{-5}{\sqrt{(-5)^2 + 144}} = \frac{-5}{13} = -\frac{5}{13}.$$

52 Use the chain rule to differentiate $f(x) = \sqrt{x^2 + 144}$, then find $f'(-9)$.

- a. $-\frac{3}{5}$ ✓
- b. $-\frac{5}{3}$
- c. $\frac{3}{5}$
- d. $-\frac{3}{4}$

$$f'(x) = \frac{x}{\sqrt{x^2 + 144}}, \text{ so } f'(-9) = \frac{-9}{\sqrt{(-9)^2 + 144}} = \frac{-9}{15} = -\frac{3}{5}.$$

53 Use the chain rule to differentiate $f(x) = \sqrt{x^2 + 16}$, then find $f'(3)$.

a. $-\frac{3}{5}$

b. $\frac{3}{4}$

c. $\frac{3}{5}$ ✓

d. $\frac{5}{3}$

$$f'(x) = \frac{x}{\sqrt{x^2 + 16}}, \text{ so } f'(3) = \frac{3}{\sqrt{(3)^2 + 16}} = \frac{3}{5} = \frac{3}{5}.$$

54 Given $h(2) = -3$, $h'(2) = -3$, $g(-3) = -1$, $g'(-3) = -4$, and $f'(-1) = -3$, find $k'(2)$ if $k(x) = f(g(h(x)))$.

a. -36 ✓

b. -10

c. 12

d. 9

$$k'(x) = f'(g(h(x))) \cdot g'(h(x)) \cdot h'(x), \text{ so}$$

$$k'(2) = f'(g(-3)) \cdot g'(-3) \cdot h'(2) = f'(-1) \cdot (-4) \cdot (-3) = (-3)(-4)(-3) = -36.$$

55 Given $h(7) = 3$, $h'(7) = -2$, $g(3) = -5$, $g'(3) = -2$, and $f'(-5) = 2$, find $k'(7)$ if $k(x) = f(g(h(x)))$.

a. 13

b. 8 ✓

c. -2

d. -4

$$k'(x) = f'(g(h(x))) \cdot g'(h(x)) \cdot h'(x), \text{ so}$$

$$k'(7) = f'(g(3)) \cdot g'(3) \cdot h'(7) = f'(-5) \cdot (-2) \cdot (-2) = (2)(-2)(-2) = 8.$$

56 Given $h(5) = 0$, $h'(5) = -5$, $g(0) = -5$, $g'(0) = 3$, and $f'(-5) = -4$, find $k'(5)$ if $k(x) = f(g(h(x)))$.

a. 60 ✓

b. -12

c. -6

d. 20

$k'(x) = f'(g(h(x))) \cdot g'(h(x)) \cdot h'(x)$, so

$$k'(5) = f'(g(0)) \cdot g'(0) \cdot h'(5) = f'(-5) \cdot (3) \cdot (-5) = (-4)(3)(-5) = 60.$$

57 Given $h(3) = 3$, $h'(3) = -3$, $g(3) = -4$, $g'(3) = -2$, and $f'(-4) = 5$, find $k'(3)$ if $k(x) = f(g(h(x)))$.

a. -15

b. 0

c. -10

d. 30 ✓

$k'(x) = f'(g(h(x))) \cdot g'(h(x)) \cdot h'(x)$, so

$$k'(3) = f'(g(3)) \cdot g'(3) \cdot h'(3) = f'(-4) \cdot (-2) \cdot (-3) = (5)(-2)(-3) = 30.$$

58 Given $h(2) = 0$, $h'(2) = 2$, $g(0) = 1$, $g'(0) = -4$, and $f'(1) = -5$, find $k'(2)$ if $k(x) = f(g(h(x)))$.

a. 40 ✓

b. -10

c. 20

d. -7

$k'(x) = f'(g(h(x))) \cdot g'(h(x)) \cdot h'(x)$, so

$$k'(2) = f'(g(0)) \cdot g'(0) \cdot h'(2) = f'(1) \cdot (-4) \cdot (2) = (-5)(-4)(2) = 40.$$

59 Which is the correct derivative of $F(x) = \sin(5x + 2)$?

- a. $5\cos(5x + 2)$ ✓
- b. $-5\sin(5x + 2)$
- c. $5\cos(x)$
- d. $\cos(5x + 2)$

By the chain rule, $F'(x) = \cos(u) \cdot u'$ where $u = 5x + 2$ and $u' = 5$: $F'(x) = 5\cos(5x + 2)$.

60 Which is the correct derivative of $F(x) = \sin(-2x - 1)$?

- a. $-2\cos(-2x - 1)$ ✓
- b. $2\sin(-2x - 1)$
- c. $-2\cos(x)$
- d. $\cos(-2x - 1)$

By the chain rule, $F'(x) = \cos(u) \cdot u'$ where $u = -2x - 1$ and $u' = -2$: $F'(x) = -2\cos(-2x - 1)$.

61 Which is the correct derivative of $F(x) = \sin(-6x)$?

- a. $6\sin(-6x)$
- b. $\cos(-6x)$
- c. $-6\cos(x)$
- d. $-6\cos(-6x)$ ✓

By the chain rule, $F'(x) = \cos(u) \cdot u'$ where $u = -6x$ and $u' = -6$: $F'(x) = -6\cos(-6x)$.

62 Which is the correct derivative of $F(x) = \sin(-6x + 4)$?

- a. $-6\cos(x)$
- b. $-6\cos(-6x + 4)$ ✓
- c. $\cos(-6x + 4)$
- d. $6\sin(-6x + 4)$

By the chain rule, $F'(x) = \cos(u) \cdot u'$ where $u = -6x + 4$ and $u' = -6$: $F'(x) = -6\cos(-6x + 4)$.

63 Which is the correct derivative of $F(x) = \sin(-5x - 2)$?

a. $\cos(-5x - 2)$

b. $-5\cos(x)$

c. $-5\cos(-5x - 2)$ ✓

d. $5\sin(-5x - 2)$

By the chain rule, $F'(x) = \cos(u) \cdot u'$ where $u = -5x - 2$ and $u' = -5$: $F'(x) = -5\cos(-5x - 2)$.